Ömer Can Atlı

Computer Engineer | Software Developer | AI & Backend

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Final-year Computer Engineering student at Gebze Technical University with a deep interest in backend systems and artificial intelligence. I build across the stack — .NET and Node.js backends, React Native and Flutter clients, and applied computer-vision and graph pipelines in Python. Creator and lead developer of two shipped products, and eight open-source projects published with the measurements that say where each one breaks.

WORK EXPERIENCE

Intern — Network Management Directorate July 2025 – August 2025

Türk Telekom, İstanbul

- Rotational internship across the Transmission, MPLS-IP, DSL, Mobile Core, Central Office and TTVPN units.
- Hands-on with xDSL/FTTH access technologies, MPLS, BNG, traffic monitoring and fault analysis on a national backbone network.

PRODUCTS

Shorties shorties.tr Creator & Lead Developer

- An exam-preparation ecosystem that turns a question bank into short vertical videos with AI: one shared architecture across three separate apps.
- Designed and built the .NET backend on a Clean Architecture split (API / Business / Core / DataAccess), the React Native (Expo) client and the Python video-generation engine, over MSSQL.
- YKS Shorties is live on the App Store and Google Play at version 2, with subscriptions through RevenueCat; KPSS and ALES are in development on the same stack.

C# · .NET · Clean Architecture · REST API · MSSQL · React Native (Expo) · Python · RevenueCat

respos resposapp.com Creator & Lead Developer

- A multi-tenant SaaS point-of-sale system for restaurants: one Node.js/Express backend serves many tenants, each with isolated data and self-service sign-up.
- Layered backend (routes → services → repositories) with tenant scoping in the service layer, JWT authentication and per-restaurant module access.
- Live table plan, kitchen display system, recipe-based stock tracking, shift and cash management, and discount authorisation with an audit log.

Node.js · Express · React · React Native (Expo) · JWT · Multi-tenancy

OPEN-SOURCE PROJECTS — github.com/Byomer1021

otonomarac · live demo Python · PyTorch · OpenCV · Gradio

Monocular driving perception and bird's-eye-view mapping

- Detection (YOLO), multi-object tracking (ByteTrack), monocular depth (Depth Anything V2), drivable-area segmentation, ground-plane homography and time-to-collision, built end to end over eight weeks.
- Deployed as a public Gradio demo on Hugging Face Spaces, running CPU-only on the free tier.

trafikisaret Python · YOLO · Dataset construction

Turkish traffic-sign dataset and two-stage recogniser

- 717 bounding boxes labelled by hand over 787 frames of own dashcam footage in daylight, rain and night.
- Class-agnostic detector plus a 14-class crop classifier: 0.953 recall on a held-out test set; detector mAP50 0.515 overall — 0.608 in daylight, 0.361 in rain and at night.

smart-city-traffic-analysis PySpark · GraphFrames · NetworkX · Pandas

38.3 million New York taxi trips modelled as a graph

- PageRank, Louvain communities and a node-removal simulation over 258 zones and 9,990 edges; full pipeline runs in 40.7 s.
- Removing the five most critical zones costs 34.1% of network flow; removing JFK Airport splits the graph from one connected component into sixteen.

plakatanima Python · PyTorch · CTC · ONNX · TensorRT · C++

Turkish licence plate recognition, measured end to end

- Reads 18 of 34 plates correctly on a held-out test set fully automatically, and 22 with the corners marked by hand — the page leads with both, because only the first is what a camera would deliver. The four-plate gap is not resolved by this sample (McNemar p ~ 0.34).
- Synthetic generator calibrated against 272 hand-labelled real plates, a CTC recogniser fine-tuned on them, and a constrained beam search over the Turkish plate grammar written in C++.
- Exported to ONNX and TensorRT FP16 on a rented T4: 24.7x on YOLO and 5.4x on the recogniser, with all 157 crops decoding to identical plates.

payflow · **live demo** Python · Recursive descent · Type systems · Pyodide

A domain-specific programming language for subscription billing

- Hand-written lexer, recursive-descent parser, type checker and tree-walking interpreter in pure Python — no parser generator, no third-party dependencies, 50 unit tests.
- Money carries its currency in the type, a domain-specific split operator rounds per step with banker's rounding, and paywall exhaustiveness is enforced by the grammar rather than checked afterwards.
- Runs live in the browser: the same five modules loaded into Pyodide, so a visitor sees the errors the real type checker raises rather than an imitation.

floorplan-parser Python · OpenCV · scikit-image · NetworkX

Floor plans to walls, doors, windows and a room graph, no learned models

- Two-scale morphology, skeleton and probabilistic Hough lines, a quarter-arc door template and a distance-transform watershed turn a raster plan into walls, openings, rooms and a room-door graph; no neural network anywhere.
- Thresholds chosen on 62 development plans and the held-out test split measured once: wall IoU 0.913, door F1 0.904, window F1 0.830, room F1 0.770 under strict one-to-one matching.

ada-llm-gnn Python · PyTorch Geometric · LoRA · vLLM · IfcOpenShell

Wheelchair accessibility of IFC buildings: LLM-generated data, GNN agents

- A large LLM generates labelled single-storey buildings that code writes to IFC and a rule engine verifies; a LoRA-tuned Qwen2.5-7B takes over the generation (81% valid against 4% untuned, 2,013 buildings).
- Three GNN agents, the route agent on a transit graph, check doors, rooms and reachability; on 336 unseen buildings the verdict reaches macro F1 0.938, and with 30% of the values deleted 3 of 225 inaccessible buildings are missed.

EDUCATION

Gebze Technical University Expected 2027 · Kocaeli

B.Sc. Computer Engineering

Şehit Mustafa Serin Science High School Graduated 2019 · Balıkesir

Science track

TECHNICAL SKILLS

Languages	Java, C++, C, C#, Python, JavaScript, SQL, HTML/CSS
Frameworks & Libraries	.NET, Spring Boot, Django, Flutter, React Native, Scikit-learn, TensorFlow (basic), NumPy, Pandas
Tools & Platforms	Git, Docker, Jira, Maven, Firebase, MSSQL, MySQL, SQL Server
Core Concepts	Object-Oriented Programming, Data Structures, Machine Learning, Deep Learning

LANGUAGES

Turkish — native · English — Advanced (C1)

VOLUNTEER WORK & EXTRACURRICULARS

C developer, "Besleme Kahramanları" · Member, GTÜ Bilgisayar Topluluğu · AFAD disaster response volunteer · Member, GTU Search and Rescue Club (GETAK)